

HBS fine fragmentation

iHighBurnupStructureFragmentation = 1–4

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Introduction: what fragmentation means (NEA)

- * few μm – 100 μm
- * high burnup, HBS pores
- * HBS fragments are acicular, central precipitation rings fragment too

burnup threshold	60 – 70 MWd/kgU (segment), significant dispersal above ~ 80
temperature threshold	$\sim 750^\circ\text{C}$ (EPRI), 640°C (NFIR), visible at $\sim 900^\circ\text{C}$ (Halden)
hydrostatic pressure	$P_h \gtrsim 40\text{--}60$ MPa suppresses release at 1500°C (NFIR)
cladding strain	5 – 10 % needed for visible fine fragmentation
heating rate	more pulverisation at 20 than at 0.2°C/s
pore pressure	65 – 80 MPa at 377°C (HBS pores)
lower size limit	a $40\mu\text{m}$ HBS fragment is stable at 1300°C

State of the art: J20 verdict on the criteria

criterion	P_g^{cr}	verdict
Gruber 1973	$P_s + (\sigma_g b + P_h F_2) / F_1$	discarded, wrong F_1, F_2
Olander 1997	$P_s + [\sigma^{cr}(1 - \varphi_2) - \sigma_\infty] / \varphi_2$	suitable, valid for HBS ($\varphi_2 \rightarrow \varphi_3$)
DiMelfi–Deitrich 1979	$P_h + G \sin \theta / [2r(1 - \cos \theta)]$	discarded
Harwell	$P_h + P_s + \frac{1}{F_3} \sqrt{\pi EG / [(1 - \nu^2)r]}$	unsuitable, no φ_2 dependence
Chakraborty–Tonks–Pastore	same with $F_4(\varphi_2)$	suitable, $\theta = 50^\circ$ only
Likhanskii–Matveev	Boyle / VdW	discarded
Worledge improved	energy balance	suitable, unphysical for $\varphi_2 < 0.25$
Irwin 1957	$P_h + \frac{1}{2} \sqrt{\pi EG / [(1 - \nu^2)r]}$	comparable with Harwell / CTP

stress criteria: small pores break first, LEFM: large pores first;

Open gaps

1. pre-transient pressure not computed by a growth model (K15: punching, OHP: from data, J19: empirical)
2. pore distribution used only by K15
3. small pores first (stress) vs large pores first (LEFM): no data
4. no criterion for the lower size limit of fragmentation
5. no direct heating-rate dependence, no pore interaction after a break

What SCIANTIX provides

ξ , $\bar{R}$, V_p , N_p , $\bar{n}$ = Xe atoms per pore, $\text{Var}(n)$, α , T , P_h

- not available: joint distribution of (n, V) , so the pressure of each pore is not known
- end of NFIR base irradiation: $\xi = 10.5 \%$, $\bar{R} = 0.35 \mu\text{m}$, $N_p = 5.8 \times 10^{17} \text{ m}^{-3}$
- measured: $\xi = 11 \%$, $\bar{R} = 0.88 \mu\text{m}$, $N_p = 4 \times 10^{16} \text{ m}^{-3}$

Pore state and pressure

Hard-sphere EoS of the porosity model, evaluated on SCIANTIX n and V_p

$$p_g = \frac{nk_B T}{V_p} Z(\eta), \quad Z = \frac{1 + \eta + \eta^2 - \eta^3}{(1 - \eta)^3}, \quad \eta = \min\left(\frac{nv_{Xe}(T)}{V_p}, 0.65\right)$$

Overpressure

$$\Delta p = p_g - P_s - P_h, \quad P_s = \frac{2\gamma}{R}, \quad \gamma = 1.1 \text{ N/m}, \quad P_h = -\sigma_h$$

Pore classes (options 3, 4): log-normal in R with the width from $\text{Var}(n)$

$$\text{CV}_R = \frac{\sqrt{\text{Var}(n)}}{3\bar{n}}, \quad \sigma_{\ln R} = \sqrt{\ln(1 + \text{CV}_R^2)}$$

Output: D^* and G^*

Each criterion returns, at every step

$$D^* = \sum_i w_i [\text{class } i \text{ broken}] \quad \text{fraction of broken pores}$$

$$G^* = \frac{\sum_i w_i n_i [\text{class } i \text{ broken}]}{\sum_i w_i n_i} \quad \text{fraction of the pore gas in broken pores}$$

w_i number weight, n_i Xe atoms of class i .

If one class, $D^* = G^* = 1$ or 0.

$$D \leftarrow \max(D, D^*), \quad G \leftarrow \max(G, G^*)$$

D : drives the fragment size G : drives the release

New HBS material and venting

α : restructured volume fraction. New material is intact

$$\alpha > \alpha_{old} : \quad D \leftarrow D \frac{\alpha_{old}}{\alpha}, \quad G \leftarrow G \frac{\alpha_{old}}{\alpha}$$

A broken pore loses its gas and stays as a void. When G grows to G_{new} , $f = \frac{1 - G_{new}}{1 - G}$

$$A \rightarrow fA, \quad B \rightarrow f^2 B, \quad \bar{n} \rightarrow f \bar{n}, \quad \text{Var}(n) \rightarrow f^2 \text{Var}(n), \quad S \rightarrow fS$$

$A = \text{Xe}$ in HBS pores, B its variance, $S = \text{HBS gas retention fraction}$. The criteria use the state before venting, $\bar{n}/S$.

Fragment size (Wigner–Seitz cell of the pores)

$$R_{WS} = \left(\frac{3}{4\pi N_p} \right)^{1/3}, \quad d = \min \left(\frac{2R_{WS}}{D}, 500 \mu\text{m} \right)$$

$$P_{cr} = P_s + \frac{\sigma_{hbs}^{cr}(1 - \xi) + P_h}{\xi}, \quad \sigma_{hbs}^{cr} = 21 \text{ MPa}$$

$$p_g \geq P_{cr} \Rightarrow D^* = G^* = 1$$

Source

J19 Eq. 22 (= J20 Eq. 7, Olander 1997)

Pressure

uniform: one mean pore; p_g from the SCIANTIX hard-sphere EoS with $\bar{n}$, V_p (no $\text{Var}(n)$)

J20 review

suitable (Olander), valid for HBS pores with $\varphi_2 \rightarrow \varphi_3$; used in SCANAIR, FALCON, RANNS

Option 2 — Kulacsy (2015)

Classes at the punching pressure before the transient, T_{ss} , $P_{h,ss}$ last cycle

$$p_0(r) = P_{h,ss} + \frac{2\gamma_{ss}}{r} + \frac{Gb}{r}, \quad G = \frac{E_{ss}}{2(1+\nu)}, \quad b = 0.39 \text{ nm}$$

$$\sigma_t = p_i - \left(\frac{3}{2} P_h + \frac{2\gamma(T, \xi)}{r} \right), \quad \sigma_f = 170 e^{-191.34 / \min(T, 1000)} \sqrt{1 - 2.62 \xi} \text{ MPa}$$

$$\ln r \sim \mathcal{N}(-0.5, 0.356^2) \text{ (} r \text{ } \mu\text{m)}, \quad D^* = \sum_i w_i [\sigma_{t,i} > \sigma_f]$$

Source	K15 Eq. 3, 7, 11, 15; own: transient start at the first fall of the fission rate, van der Waals in place of Ronchi's EoS
Pressure	not uniform, $\Delta p \sim Gb/r$ (small pores more loaded); gas per class from van der Waals at T_{ss} , not from SCIANTIX n , $\text{Var}(n)$ (only ξ , T , P_h enter)
J20 review	used by Kulacsy with $F_1 = 1$ instead of the correct $\frac{1}{2}$ for a spherical pore

Option 3 — LEFM (Irwin)

Penny-shaped crack of radius R

$$P_{cr}(R) = P_h + \frac{2\gamma}{R} + \frac{1}{2Y} \sqrt{\frac{\pi E G_{hbs}}{(1-\nu^2)R}}, \quad G_{hbs} = 0.055 \text{ J/m}^2 \text{ (fitted), } Y = 1$$

Pores with $R \geq R_c$, $P_{cr}(R_c) = p_g$, break; $z = \ln(R_c/\bar{R})/\sigma_{\ln R}$

$$D^* = \frac{1}{2} \operatorname{erfc}\left(\frac{z}{\sqrt{2}}\right), \quad G^* = \frac{1}{2} \operatorname{erfc}\left(\frac{z - 3\sigma_{\ln R}}{\sqrt{2}}\right)$$

Source	Irwin (1957), J20 Eq. 8; own: capillary term, G_{hbs} fitted on NFIR, log-normal classes
Pressure	uniform over the classes, p_g from the SCIANTIX EoS with $\bar{n}$; $\operatorname{Var}(n)$ only gives the class width
J20 review	comparable with Harwell and CTP; bulk G (2 J/m ²) is ~ 2 orders too strong, local G is missing

Option 4 — J19 force balance per class, progressive rupture

Pressure per class, level from SCIANTIX gas

$$p_i = P_h + \frac{2\gamma}{R_i} + \kappa \frac{c_{dp}}{R_i}, \quad \sum_i w_i \text{EoS}^{-1}(p_i; V_i, T) = \bar{n}, \quad c_{dp} = 55 \text{ N/m}$$

Rupture per class and percolation

$$p_i \geq P_{cr,i} = \frac{2\gamma}{R_i} + \frac{\sigma_{hbs}^{cr}(1 - \xi) + P_h}{\xi}, \quad \xi \geq 0.29 \Rightarrow D^* = G^* = 1$$

Progressive rupture

$$D \leftarrow D + (D^* - D)(1 - e^{-\Delta t/\tau}), \quad \tau = 2 \text{ s}$$

Source	J19 Eq. 21–22, Table 4; own: κ mass balance, class sum, relaxation (as GrainBoundaryMicroCracking.C); τ from the J19 microcracking mode
Pressure	per class, $\Delta p \sim 1/R$; level from the SCIANTIX EoS and $\bar{n}$, class width from $\text{Var}(n)$; no fitted parameter
J20 review	same criterion as option 1 (Olander); J19 rupture is instantaneous

Periodic RVE, 1200 spherical pores, linear elasticity, zero macroscopic strain; pore i breaks at

$$\sigma_l^{max}|_i > \sigma_R = 275 \text{ MPa}, \quad \hat{s}_i = \frac{\sigma_l^{max}|_i}{p_{in}} \in [0.6, 1.5] \quad (\text{isolated pore: } 0.5)$$

Same structure as J19, with the stress factor $\hat{s}$ in place of $\xi/(1 - \xi)$

$$\Delta p_{c,i} = \frac{\sigma_R + P_h}{\hat{s}_i} \quad \text{vs} \quad \Delta p_c = \frac{\sigma_{hbs}^{cr} + P_h}{\xi/(1 - \xi)}$$

Linear elasticity: stress on pore i from the pressure of all pores, influence coefficients S_{ij} of the RVE

$$\sigma_i = \sum_j S_{ij} (p_j - P_h) \Rightarrow \sigma_i = \hat{s}_i (p_i - P_h), \quad \hat{s}_i = S_{ii} + \sum_{j \neq i} S_{ij} \frac{p_j - P_h}{p_i - P_h}$$

Regression of $\hat{s}_i$ on what SCIANTIX knows (ξ, N_p, R_i) , the neighbours giving the scatter

$$\ln \hat{s}_i = a + b \ln \frac{R_i}{\bar{R}} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (a, b, \sigma_\varepsilon) = f(\xi, \sigma_{\ln R}) \text{ from the RVE}$$

Pressure per class from the SCIANTIX gas.

Propagation 1 and 2: cascade and percolation

1. **Sequential breaking** (RVE, no new FE run: S_{ik} from one load case per pore). Break pore $k = \arg \max_i \sigma_i / \sigma_R$, then $p_k \rightarrow P_h$

$$\sigma_i^{(m+1)} = \sigma_i^{(m)} - S_{ik} (p_k - P_h) + \Delta \sigma_i^{\text{void}}, \quad \text{repeat while } \max_i \sigma_i / \sigma_R \geq 1$$

Output: size of the cascade at fixed p_{in} ; reduced form $\hat{\sigma}_{eff} = \hat{\sigma} (1 + \chi(\xi) D)$, χ from the RVE, sign of $\Delta \sigma^{\text{void}}$ to be computed.

2. **Release by connectivity** (bond percolation). Pores = nodes, ligaments = bonds;

bond ij failed if $\sigma_{lig,ij} \geq \sigma_c$

$$G = \frac{\sum_{i \in \mathcal{C}_\partial} V_i n_i}{\sum_i V_i n_i}, \quad \mathcal{C}_\partial = \text{pores connected to the free surface through failed bonds}$$

$$G = \Phi_G(D, \xi), \quad d = \Phi_d(D, \xi) = \text{size of the intact clusters enclosed by the failed bonds}$$

Replaces $G^* =$ gas of the broken pores and $A_f = 2\pi R_{WS}^2$ [?]

Propagation 3: crack from a pore with gas expansion

Penny crack of radius a opened by the pressure p (Sneddon); the gas of the pore (n fixed) fills the new volume

$$K_I = 2p\sqrt{a/\pi}, \quad V_c = \frac{16(1-\nu^2)}{3E} p a^3, \quad \mathcal{G} = \frac{K_I^2(1-\nu^2)}{E} = \frac{4(1-\nu^2)}{\pi E} p^2 a$$

$$p = \text{EoS}(n, V_p + V_c(p, a), T), \quad \text{growth if } \mathcal{G}(a) \geq \mathcal{G}_c$$

For $V_c \gg V_p$ (ideal gas): $p \propto a^{-3/2}$, hence $\mathcal{G} \propto a^{-2}$: the crack arrests

$$a_{arr} : \mathcal{G}(p(a_{arr}), a_{arr}) = \mathcal{G}_c$$

Link between pores i, j at distance ℓ_{ij} , to be checked on the RVE with a crack path (cohesive zone / XFEM)

$$a_{arr,i} + a_{arr,j} \geq \ell_{ij} - R_i - R_j \Rightarrow \text{bond } ij \text{ failed}$$

Explains pores that do not burst and the stable 40 μm fragments (NEA); one parameter, $\mathcal{G}_c$.

What returns to SCIANTIX, and how

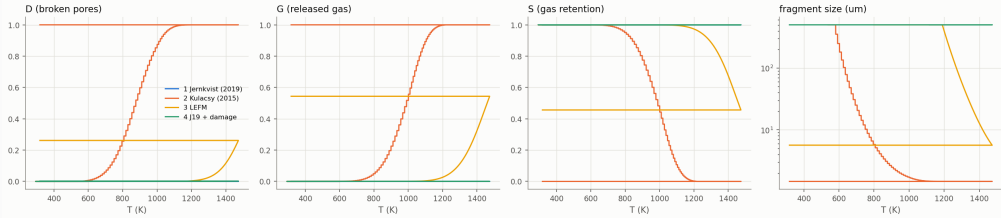
Offline: MMM on a grid $\xi \in [0.05, 0.25]$, $\sigma_{\ln R} \in [0, 0.5]$, 10 seeds each; fits; tables read by SCIANTIX at start (as the other tabulated data); no FE at run time.

from the RVE / MMM	in SCIANTIX	interpolation	replaces
$a, b, \sigma_\varepsilon (\xi, \sigma_{\ln R})$	$\hat{s}_i$ of each class	2D table	ligament factor $\xi/(1-\xi)$, strength spread
$\chi(\xi)$	$\hat{s}_{eff}(D)$	1D table	nothing (no interaction)
$\Phi_G(D, \xi)$	release G from D	2D table	G^* gas-weighted, venting at $\xi = 0.29$
$\Phi_d(D, \xi)$	fragment size d	2D table	$A_f = 2\pi R_{WS}^2$
$\mathcal{G}_c, a_{arr}$ rule	bond criterion	closed form	uniform-pressure LEFM, G_{hbs}

Input from SCIANTIX each step: $\xi, N_p, \bar{R}, \sigma_{\ln R}$ (from $\text{Var}(n)$), $p_i = \text{EoS}(n_i, V_i, T)$, T, P_h .

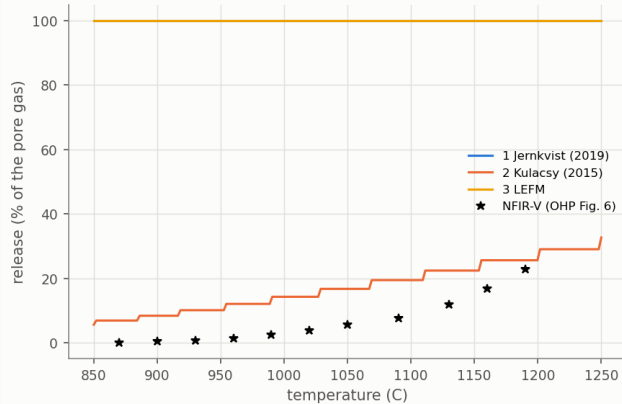
Checks: reduced model against the RVE FGR–pressure curve (OHP Fig. 7), then NFIR, P_h scaling, IFA-650.10 control.

NFIR anneal

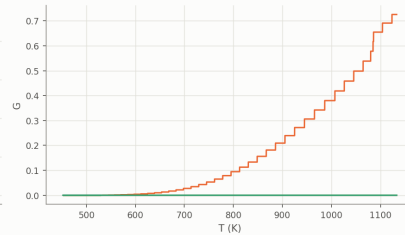
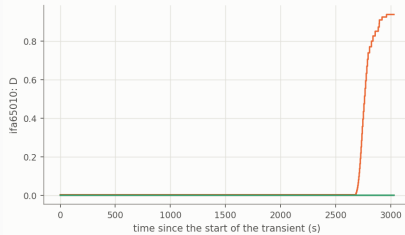
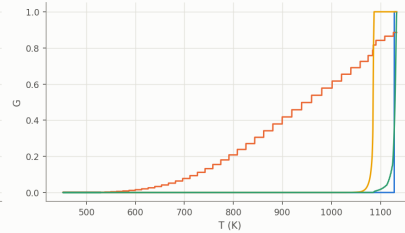
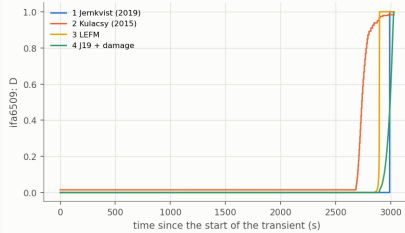


release at 1200 °C (data 29.8 %): 1: 0 2: 56.5 % 3: 30.7 % 4: 0.02 %

NFIR: Kr-85 release, prescribed state



IFA-650.9 and .10



	1	2	3	4
NFIR, release at 1200 °C	0	56.5 %	30.7 %	0.02 %
IFA-650.9	$D = 1$	$D = 0.98$	$D = 1$	$D = 1$
IFA-650.10 (control)	0	$D = 0.94$	0	0